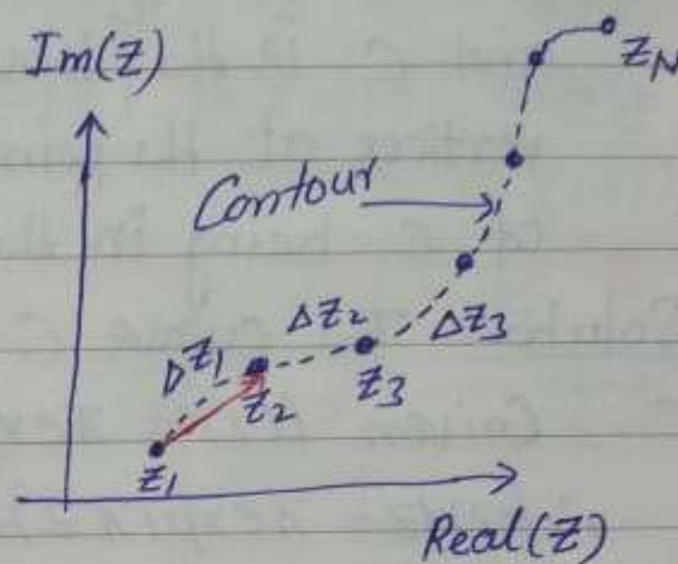
Module-4
Complex Integration

1. Integration of a complex function along a contour
2. Cauchy-Goursat theorem
3. Cauchy’s integral formula
4. Cauchy’s residue theorem
5. Evaluation of real integrals
6. Indented contour integral

Contour Integration

For real case $\int_a^b f(x)dx$

For complex case $\int_{z_1}^{z_N} f(z)dz$



suppose end points z_1 and z_N are fix and take $\lim N \to \infty$ and hence $\Delta z_n \to 0$.

Such trajectory is called contour and denoted by a symbol Γ, Hence contour integral over Γ is

$$\int_C f(z)dz \text{ or } \int_\Gamma f(z)dz = \lim_{N \to \infty} \sum_{N=1}^{N-1} \Delta z_n f(z_n)$$

Contour integral along a parametric curve

When the trajectory Γ or C is

$z = z(t)$ $(a \le t \le b)$ where $z_1 = z(a)$ & $z_2 = z(b)$

Suppose that $f(z(t))$ is piecewise continuous in the interval $a \le t \le b$. Then

$$\boxed{\int_C f(z)\,dz = \int_a^b f[z(t)]\,z'(t)\,dt}$$

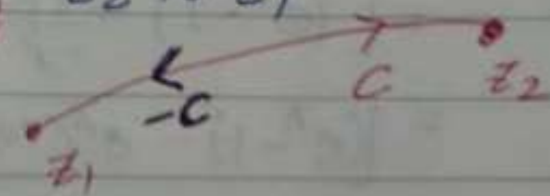
$z'(t)$ piecewise continuous on $a \le t \le b$

Properties (i) $\boxed{\int_C [f(z)+g(z)]\,dz = \int_C f(z)\,dz + \int_C g(z)\,dz}$

(ii) Reverse order: C is from $z_1 \xrightarrow{to} z_2$

$-C$ is from z_2 to z_1

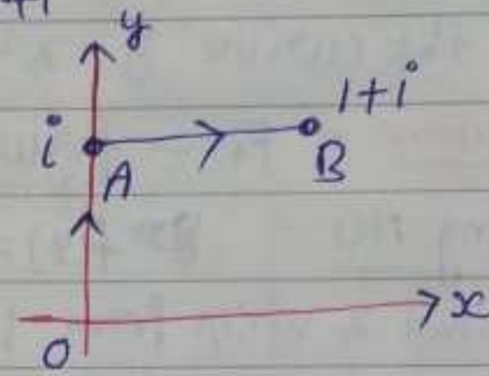
$$\therefore \boxed{\int_{-C} f(z)\,dz = -\int_C f(z)\,dz}$$



(iii) If $C = C_1 + C_2$ then

$$\boxed{\int_C f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz}$$

Ex-3 Evaluate $\int_C f(z)dz$, where $f(z) = y - x - 3ix^2$ and C consists of line segments $z=0$ to $z=i$ and other form $z=i$ to $z=1+i$



Solution $\int_C f(z)dz = \int_{OA} f(z)dz + \int_{AB} f(z)dz$ (1)

Along OA $x=0 \Rightarrow dx=0$ and y varies from 0 to 1

$\therefore f(z) = y - 0 - 0 = y$

$dz = dx + idy = idy$

Along AB $y = 1 \Rightarrow dy = 0$ and x varies from 0 to 1

$\therefore f(z) = 1 - x - 3ix^2$ & $dz = dx + idy = dx$

Hence from (1)

$$\int_C f(z)dz = \int_0^1 y(idy) + \int_0^1 (1 - x - 3ix^2)dx$$

$$= i\left(\frac{y^2}{2}\right)_0^1 + \left(x - \frac{x^2}{2} - ix^3\right)_0^1$$

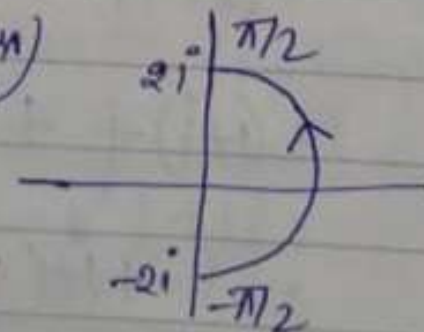
$$= \frac{i}{2} + \left(1 - \frac{1}{2} - i\right)$$

$$\boxed{\int_C f(z)dz = \frac{1-i}{2}}$$

Ex 4 Evaluate $\int_C f(z)\,dz$, where $f(z)=\bar{z}$ and $z=2e^{i\theta}$ with $\left(-\pi/2 \le \theta \le \pi/2\right)$

Solution: Equation of circle.

$$z = Re^{i\theta} \quad \text{or} \quad |z-a| = R \quad (R \text{ is radius})$$



Given $z = 2e^{i\theta}$, $\Rightarrow \bar{z} = 2e^{-i\theta}$

and $dz = 2ie^{i\theta}\,d\theta$

$$\therefore \int_C \bar{z}\,dz = \int_{-\pi/2}^{\pi/2}\left(2e^{-i\theta}\right)\left(2ie^{i\theta}\,d\theta\right)$$

$$= 4i\int_{-\pi/2}^{\pi/2} d\theta = 4i\left(\pi/2+\pi/2\right) = 4\pi i$$

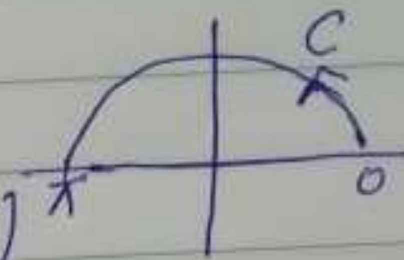
Ex-5 Evaluate $\int_C f(z)dz$ where $f(z)=\dfrac{z+2}{z}$ and C is

(a) semi circle $z=2e^{i\theta}$ $(0\le\theta\le\pi)$ Ans $-4+2\pi i$

(b) semicircle $z=2e^{i\theta}$ $(\pi\le\theta\le 2\pi)$ Ans $4+2\pi i$

(c) the circle $z=2e^{i\theta}$ $(0\le\theta\le 2\pi)$ Ans $4\pi i$

Hints (a) $\int_C f(z)dz = \int_C \left(\dfrac{z+2}{z}\right)dz$



$$= \int_0^\pi \frac{(2e^{i\theta}+2)}{2e^{i\theta}}(2ie^{i\theta}d\theta)$$

$$= 2i\int_0^\pi (e^{i\theta}+1)d\theta$$

$$= 2i\left(\frac{e^{i\theta}}{i}+\theta\right)_0^\pi$$

$$= 2i\left[\frac{1}{i}(e^{\pi i}-1)+(\pi-0)\right]$$

$$= 2i\left(-\frac{2}{i}+\pi\right)$$

$$\boxed{\int_C f(z)dz = -4+2\pi i}$$

$0\le\theta\le\pi$

Integral of a Nonanalytic Function. Dependence on Path

Integrate $f(z) = \text{Re}\, z = x$ from 0 to $1 + 2i$ (a) along C^*

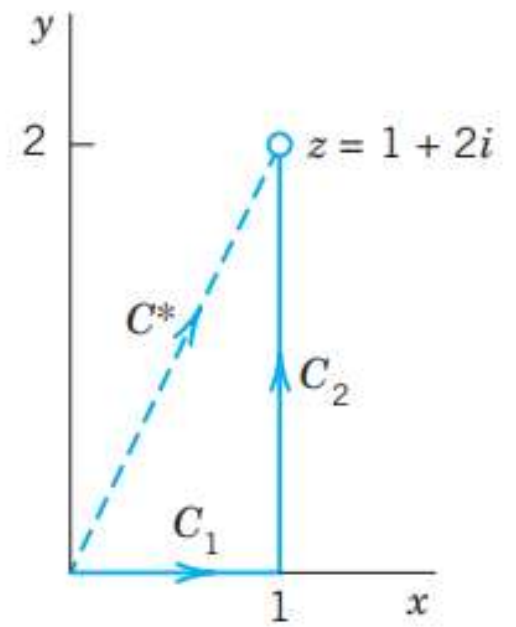
(b) along C consisting of C_1 and C_2.

Solution. **(a)** C^* can be represented by $z(t) = t + 2it \ (0 \leqq t \leqq 1)$.

Hence $\dot{z}(t) = 1 + 2i$ and $f[z(t)] = x(t) = t$ on C^*.

We now calculate

$$\int_{C^*} \text{Re}\, z\, dz = \int_0^1 t(1 + 2i)\, dt = \frac{1}{2}(1 + 2i) = \frac{1}{2} + i.$$

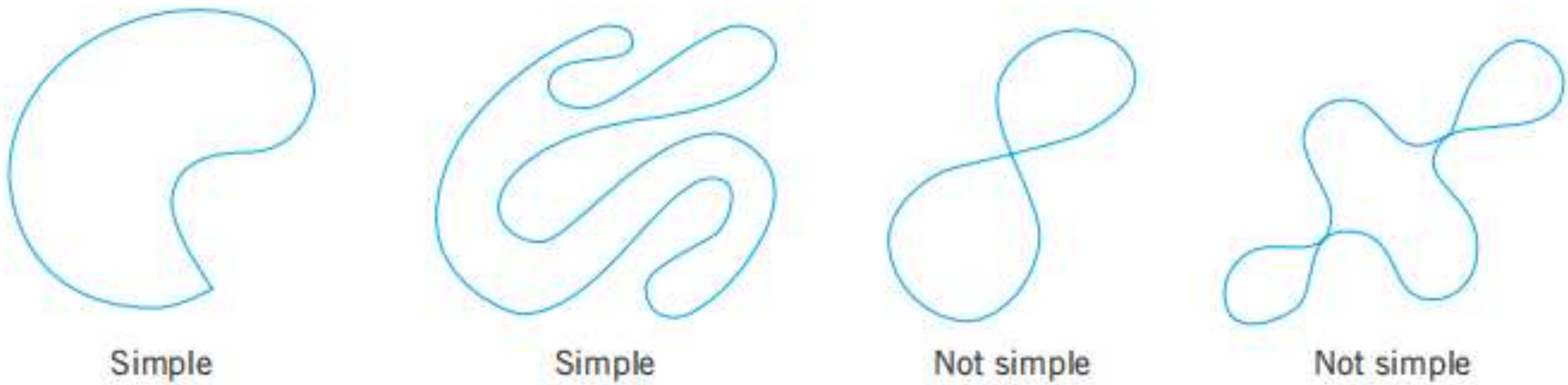


b)

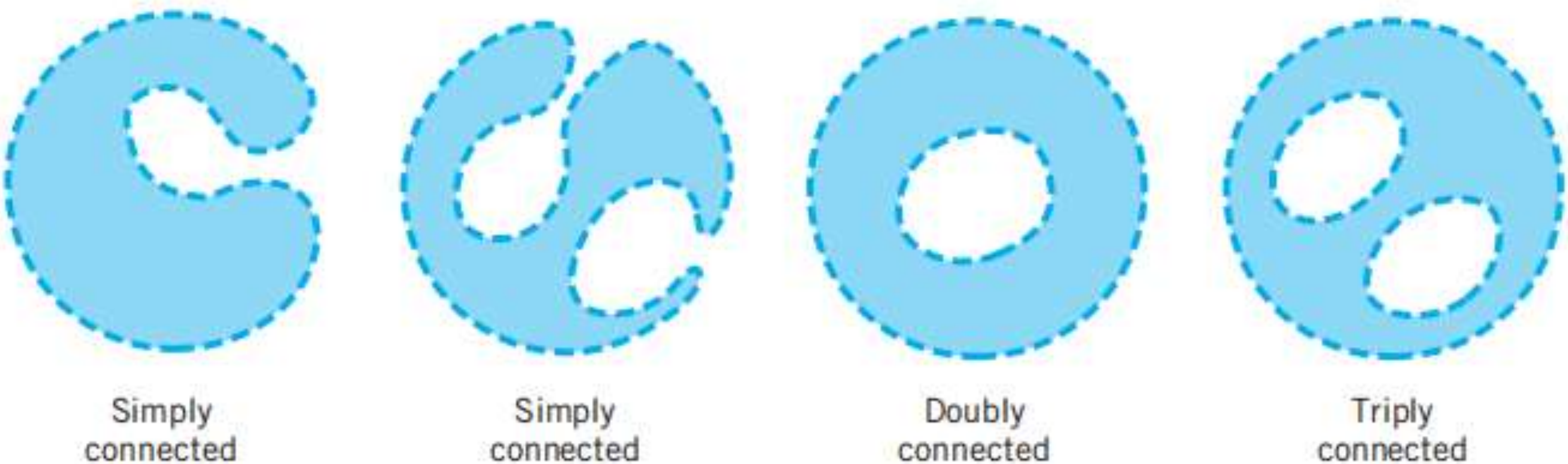
$$\int_C \text{Re}\, z\, dz = \int_{C_1} \text{Re}\, z\, dz + \int_{C_2} \text{Re}\, z\, dz$$

$$= \int_0^1 t\, dt + \int_0^2 1 \cdot i\, dt$$

$$= \frac{1}{2} + 2i.$$



Closed paths



Simply and multiply connected domains

Cauchy–Goursat Theorem

If a function f(z) is analytic at all points interior to and on a simple closed contour C, then

$$\int_C f(z)\, dz = 0.$$

Example If *C* denotes any closed contour lying in the open disk |z| < 2, t

$$\int_C \frac{z\, e^z}{(z^2+9)^5}\, dz = 0.$$

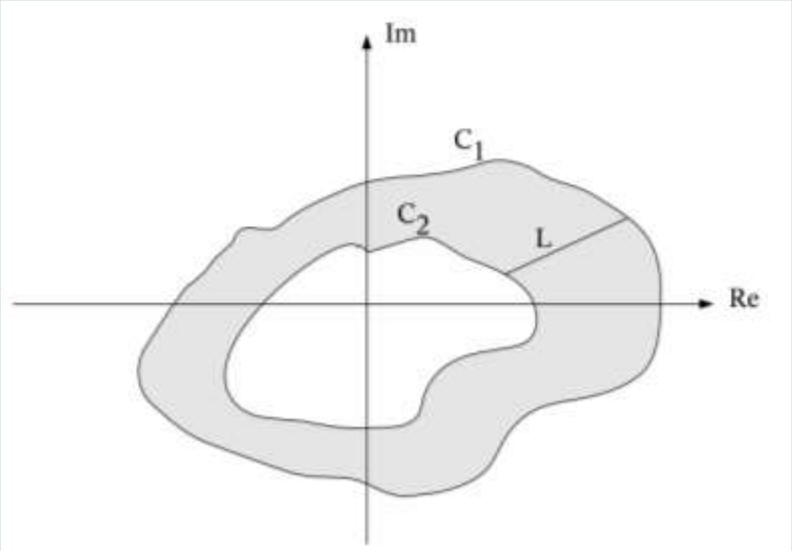
Because the disk is a simply connected domain with two singularities *z* = ± 3*i* of the integral are outside to the disk.

Corollary.

Let C_1 be a positively oriented simple closed contour and let C_2 be a positively oriented simple closed contour entirely inside the interior of C_1. If f is analytic in between and on C_1 and C_2, then

$$\int_{C_1} f(z)\,dz - \int_{C_2} f(z)\,dz = 0.$$

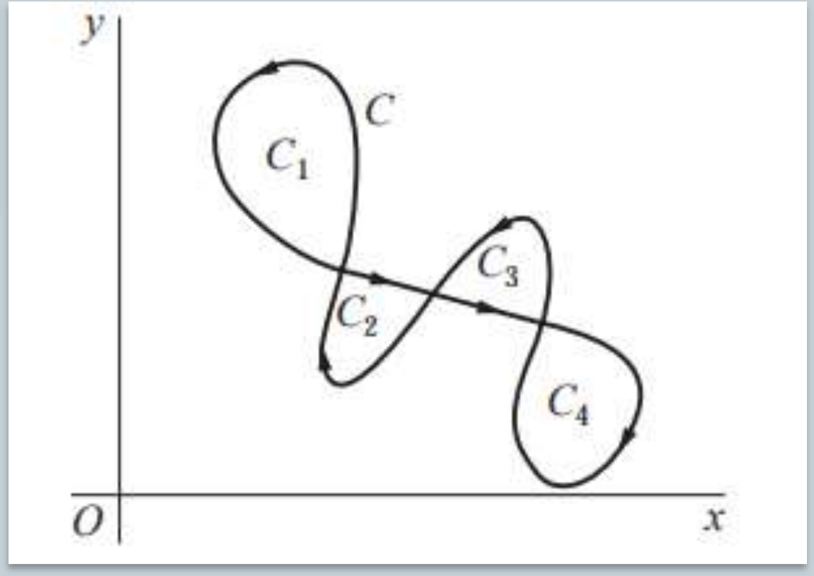
$$\boxed{\int_{C_1} f(z)\,dz = \int_{C_2} f(z)\,dz.}$$



Example

If C is a simple closed curve (but intersects itself a finite number of times with simple closed contours C_k (k = 1, 2, 3, 4) and suppose the function f (z) is analytic at each point interior to and on C. Then by Cauchy-Goursat theorem the value of the integral around each C_k is zero and

$$\int_C f(z)\,dz = \sum_{k=1}^{4} \int_{C_k} f(z)\,dz = 0.$$



Cauchy Integral formula

Suppose f(z) be analytic everywhere inside and on a simple closed contour C, taken in the positive sense. If z_0 is any point interior to C, then

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)\,dz}{z - z_0}.$$

It can be used to evaluate certain integrals along a simple closed contours.

$$\int_C \frac{f(z)\,dz}{z - z_0} = 2\pi i f(z_0),$$

Derivatives of an Analytic Function

If $f(z)$ is analytic in a domain D, then it has derivatives of all orders in D, which are then also analytic functions in D. The values of these derivatives at a point z_0 in D are given by the formulas

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz$$

$$f''(z_0) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^3} dz$$

and in general

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \qquad (n = 1, 2, \cdots);$$

Cauchy Integral formula for n^{th} derivatives

Suppose f(z) be analytic everywhere inside and on a simple closed contour C, taken in the positive sense. If z_0 is any point interior to C, then

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_C \frac{f(z)\,dz}{(z-z_0)^{n+1}} \quad (n = 1, 2, \ldots)$$

It can be used to evaluate certain integrals along a simple closed contours.

$$\int_C \frac{f(z)\,dz}{(z-z_0)^{n+1}} = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

$$(n = 1, 2, \ldots)$$

Entire Functions

$$\oint_C e^z\,dz = 0, \qquad \oint_C \cos z\,dz = 0, \qquad \oint_C z^n\,dz = 0 \qquad (n = 0, 1, \cdots)$$

for any closed path, since these functions are entire (analytic for all z).

Points Outside the Contour Where $f(x)$ is Not Analytic

$$\oint_C \sec z\,dz = 0, \qquad \oint_C \frac{dz}{z^2+4} = 0 \quad \text{where } C \text{ is the unit circle,}$$

$\sec z = 1/\cos z$ is not analytic at $z = \pm\pi/2, \pm 3\pi/2, \cdots$, but all these points lie outside C; none lies on C or inside C. Similarly for the second integral, $z = \pm 2i$ outside C.

EXAMPLE $$\oint_C \bar{z}\, dz = \int_0^{2\pi} e^{-it} i e^{it}\, dt = 2\pi i$$

where C: $z(t) = e^{it}$ is the unit circle.

EXAMPLE $$\oint_C \frac{dz}{z^2} = 0$$ where C is the unit circle.

EXAMPLE $$\oint_C \frac{dz}{z} = 2\pi i$$ where C is the unit circle.

Ex 1 Evaluate $\int_C \frac{ze^z}{(z^2+9)^5}\,dz$ in a closed contour $|z|<2$

Solution Given $f(z) = \frac{ze^z}{(z^2+9)^5}$ has ~~$z = \pm 3i$~~ singular points at $z = \pm 3i$ but these points are outside $|z|<2$

Hence $f(z)$ is analytic everywhere in $|z|<2$

$\therefore$ By Cauchy Goursat Theorem $\boxed{\int_C f(z)\,dz = \int_C \frac{ze^z\,dz}{(z^2+9)^5} = 0}$

Ex 2 Apply Cauchy Goursat Theorem and show that $\int_C f(z)\,dz = 0$ in either direction, when where C is the unit circle $|z|=1$

(i) $f(z) = \frac{z^2}{z-3}$ has singular point at $z=3$ and hence outside to $|z|=1$ $\therefore \int_C f(z)\,dz = 0$ (By G.G Theorem)

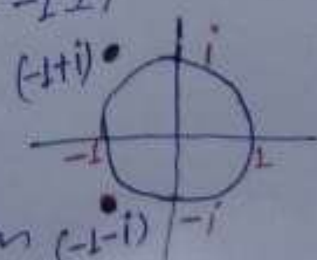
(ii) $f(z) = ze^{-z}$ has no singular point, hence analytic everywhere

Therefore by G.G. Theorem $\int_C f(z)\,dz = 0$

(iii) $f(z) = \frac{1}{z^2+2z+2}$ has two singular points at $z = -1 \pm i$ and are outside of $|z|=1$

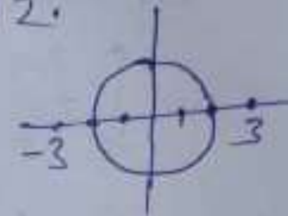
Hence analytic everywhere

$\therefore \int_C f(z)\,dz = 0$ by C.G. Theorem



Ex.3 Evaluate $\int_C \frac{z}{(9-z^2)} dz$ where C is circle $|z|=2$

Solution Given that $f(z) = \frac{z}{9-z^2}$ has singular points at $z = \pm 3$ and these points are outside of circle radius 2.



Hence by C.G. Theorem

$$\int_C \frac{z}{(9-z^2)} dz = 0$$

Ex-4 Evaluate $\int_C \frac{\exp(2z)\, dz}{z^4}$ where C is $|z|=1$

Solution Given $f(z) = \frac{e^{2z}}{z^4}$ has a singular point at $z=0$

Hence by Cauchy nth derivative formula

$$\int_C \frac{f(z)\,dz}{(z-z_0)^{n+1}} = \frac{2\pi i}{n!} f^n(z_0)$$

Hence $\int_C \frac{\exp(2z)dz}{z^4} = \int_C \frac{\exp(2z)dz}{(z-0)^{3+1}} = \frac{2\pi i}{3!} f'''(0) = \frac{2\pi i}{6} \times 8 = \frac{8\pi i}{3}$

where $f(z) = e^{2z} \Rightarrow f'''(z) = 8e^{2z}$

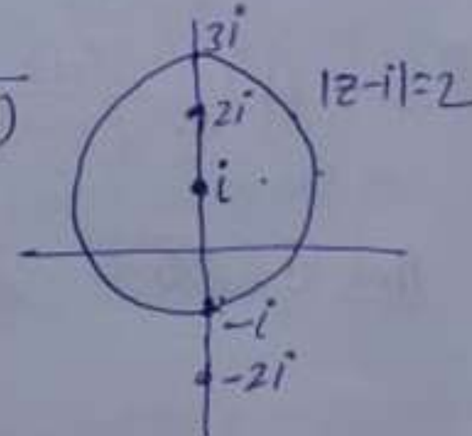
$\Rightarrow f'''(0) = 8$

$\therefore \boxed{\int_C \frac{\exp(2z)dz}{z^4} \text{ in } |z|=1 \text{ is } \frac{8\pi i}{3}}$

in the +ve

$\therefore \int_C \frac{\exp(\ldots)}{z^4} \ldots$

Ex-5 Evaluate $\int_C f(z)\,dz$ around the circle $|z-i|=2$ in the +ve sense when (i) $f(z)=\frac{1}{z^2+4}$ (ii) $f(z)=\frac{1}{(z^2+4)^2}$



Solution (i) Given that $g(z)=\frac{1}{(z^2+4)}=\frac{1}{(z+2i)(z-2i)}$

has singular points at $z=\pm 2i$

Here $\cancel{z}=-2i$ outside of $|z-i|=2$

Hence by Cauchy integral formula

$$\int_C \frac{f(z)dz}{(z-z_0)}=\int_C \frac{1/dz}{(z+2i)(z-2i)}=\int_C \frac{f(z)dz}{(z-2i)} = 2\pi i f(2i)$$

where $f(z)=\frac{1}{(z+2i)} \Rightarrow f(2i)=\frac{1}{2i+2i}=\frac{1}{4i}$

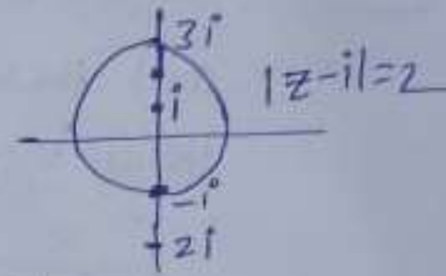
$$\therefore \int_C \frac{dz}{(z^2+4)} = 2\pi i f(2i) = 2\pi i \times \frac{1}{4i} = \pi/2$$

$$\therefore \boxed{\int_C \frac{dz}{(z^2+4)}=\frac{\pi}{2} \text{ in } |z-i|=2}$$

(ii) Given that $g(z) = \dfrac{1}{(z^2+4)^2} = \dfrac{1}{(z+2i)^2(z-2i)^2} = f(z)/(z-2i)^2$

By Cauchy nth ~~[illegible]~~ derivative formula

$$\int_c \frac{f(z)\,dz}{(z-z_0)^{n+1}} = \frac{2\pi i}{n!} f^n(z_0) \text{ where } z_0 \in C$$



$$\text{Hence } \int_{|z-i|=2} \frac{1}{(z^2+4)^2}\,dz = \int_{|z-i|=2} \frac{1}{(z+2i)^2(z-2i)^2}\,dz = \int_{|z-i|=2} \frac{f(z)\,dz}{(z-2i)^{1+1}}$$

$$= \frac{2\pi i}{1!} f'(2i)$$

$$= 2\pi i \times \cancel{4}\,\frac{1}{32i}$$

$$= \frac{\pi}{16}$$

Here $f(z) = \dfrac{1}{(z+2i)^2} \Rightarrow f'(z) = \dfrac{-2}{(z+2i)^3}$

$$\Rightarrow f'(2i) = -\frac{2}{(4i)^3}$$

$$= \frac{2}{64i}$$

$$= \frac{1}{32i}$$

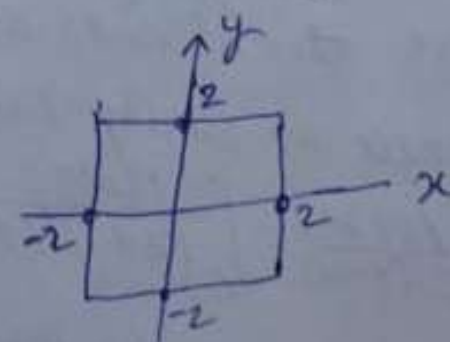
Hence $\boxed{\displaystyle\int_{|z-i|=2} \frac{dz}{(z^2+4)^2} = \frac{\pi}{16}}$

Ex-6 Evaluate $\oint$ following integrals, where C is +vely oriented boundary of the square whose sides lie along the lines $x = \pm 2$ and $y = \pm 2$

(i) $\int_C \frac{e^{-z}\,dz}{\left(z - \frac{\pi i}{2}\right)}$ (ii) $\int_C \frac{\cos z\,dz}{z(z^2+8)}$ (iii) $\int_C \frac{z\,dz}{2z+1}$ (iv) $\int_C \frac{\cosh z}{z^4}\,dz$ (v) $\int_C \frac{\tan(z/2)\,dz}{(z-x_0)^2}$

$-2 < x_0 < 2$

Ans (i) 2π, (ii) $\frac{\pi i}{4}$ (iii) $-\frac{\pi i}{2}$ (iv) 0 (v) $i\pi \sec^2(x_0/2)$



Residue

Let $f(z)$ has a Laurent series

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \frac{b_1}{z - z_0} + \frac{b_2}{(z - z_0)^2} + \cdots$$

Residue

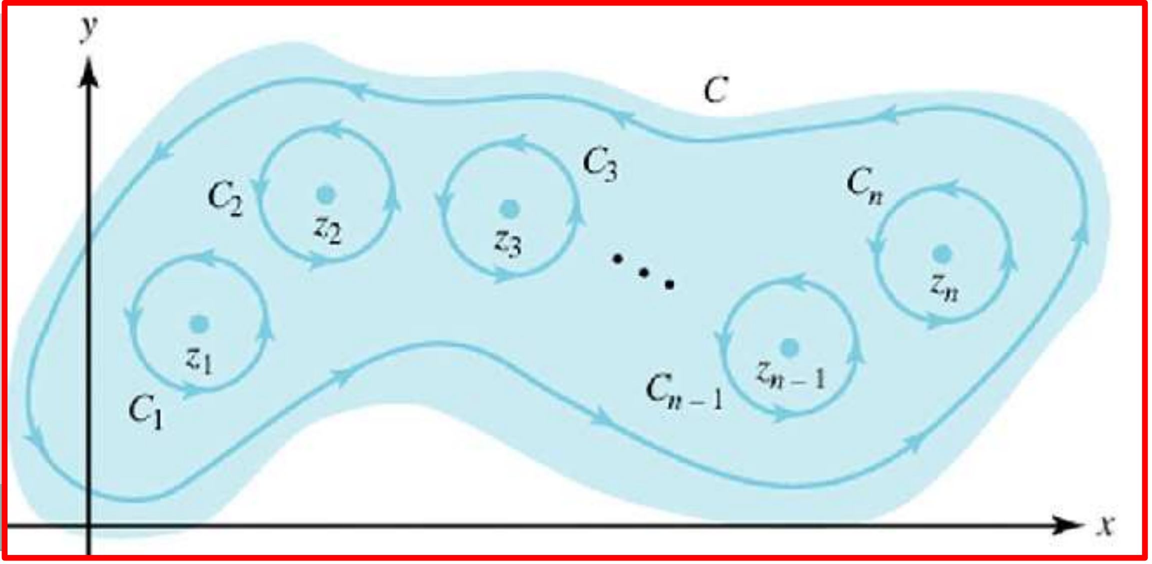
$$b_1 = \frac{1}{2\pi i} \oint_C f(z)\, dz.$$

$$\oint_C f(z)\, dz = 2\pi i b_1.$$

Residue Theorem

Let f(z) be analytic inside a simple closed path C and on C, except for finitely many singular points z_1 , z_2, ... , z_k inside C. Then the integral of f(z) taken counterclockwise around C equals $2\pi i$ times the sum of the residues of f(z) at z_1, ... , z_k

$$\oint_C f(z)\, dz = 2\pi i \sum_{j=1}^{k} \operatorname*{Res}_{z=z_j} f(z).$$



Example on Residue Theorem

Evaluate $\displaystyle\oint_{|z|=2} \frac{5z-2}{z(z-1)}\,dz.$

Write $f(z) = \dfrac{5z-2}{z(z-1)}$. For $0 < |z| < 1$,

has singular ponits at z = 0 and 1

$\text{Res}\,(f, 0) = 2$ and

$\text{Res}\,(f, 1) = 3.$

Hence, by residue theorem

$$\oint_{|z|=2} \frac{5z-2}{z(z-1)}\,dz = 2\pi i\,[\text{Res}\,(f,0) + \text{Res}\,(f,1)]$$
$$= 10\pi i.$$

$$f(z) = \frac{\sin z}{z^4} = \frac{1}{z^3} - \frac{1}{3!z} + \frac{1}{5!} - \frac{z^3}{7!} + - \cdots$$

This series shows that $f(z)$ has a pole of third order at $z = 0$ and the residue $b_1 = -\frac{1}{3}!$. Hence

$$\oint_C \frac{\sin z}{z^4}\, dz = 2\pi i b_1 = -\frac{\pi i}{3}.$$

$\boxed{\text{Integrate } f(z) = 1/(z^3 - z^4) \text{ clockwise around the circle } C\text{: } |z| = \tfrac{1}{2}.}$

$z^3 - z^4 = z^3(1 - z)$ shows that $f(z)$ is singular at $z = 0$ and $z = 1$.

Now $z = 1$ lies outside C.

the Laurent series

$$\frac{1}{z^3 - z^4} = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + 1 + z + \cdots \qquad (0 < |z| < 1).$$

Clockwise integration thus yields

$$\boxed{\oint_C \frac{dz}{z^3 - z^4} = -2\pi i \operatorname*{Res}_{z=0} f(z) = -2\pi i.}$$

But $$\frac{1}{z^3 - z^4} = -\frac{1}{z^4} - \frac{1}{z^5} - \frac{1}{z^6} - \cdots \qquad (|z| > 1)$$

Evaluate the following integral counterclockwise around any simple closed path such that C

(a) 0 and 1 are inside

(b) 0 is inside, 1 outside,

(c) 1 is inside, 0 outside,

(d) 0 and 1 are outside.

$$\oint_C \frac{4-3z}{z^2-z}\,dz$$

The integrand has simple poles at 0 and 1, with residues

$$\operatorname*{Res}_{z=0} \frac{4-3z}{z(z-1)} = \left[\frac{4-3z}{z-1}\right]_{z=0} = -4,$$

$$\operatorname*{Res}_{z=1} \frac{4-3z}{z(z-1)} = \left[\frac{4-3z}{z}\right]_{z=1} = 1.$$

Answer: (a) $2\pi i(-4+1) = -6\pi i$, (b) $-8\pi i$, (c) $2\pi i$, (d) 0

Integrate $(\tan z)/(z^2 - 1)$ counterclockwise around the circle C: $|z| = \frac{3}{2}$.

Solution. $\tan z$ is not analytic at $\pm\pi/2, \pm 3\pi/2, \cdots$,
but all these points lie outside the contour C.
the given function has simple poles at ± 1.
the denominator $z^2 - 1 = (z - 1)(z + 1)$
Hence by the residue theorem

$$\oint_C \frac{\tan z}{z^2 - 1}\, dz = 2\pi i \left(\operatorname*{Res}_{z=1} \frac{\tan z}{z^2 - 1} + \operatorname*{Res}_{z=-1} \frac{\tan z}{z^2 - 1} \right)$$

$$= 2\pi i \left(\left.\frac{\tan z}{2z}\right|_{z=1} + \left.\frac{\tan z}{2z}\right|_{z=-1} \right)$$

$$= 2\pi i \tan 1$$

Exercise

$$\oint_C \frac{e^z}{\cos z}\,dz, \quad C\colon |z - \pi i/2| = 4.5$$

Simple poles at $\pi/2$, residue $e^{\pi/2}/(-\sin \pi/2)$, and at $-\pi/2$, residue $e^{-\pi/2}/\sin \pi/2 = e^{-\pi/2}$. *Answer:* $-4\pi i \sinh \pi/2$

$$\oint_C \frac{\sinh z}{2z - i}\,dz, \quad C\colon |z - 2i| = 2$$

$$2\pi i\,(\sinh \tfrac{1}{2} i)/2 = -\pi \sin \tfrac{1}{2}$$

Theorem

If a function f(z) is analytic everywhere in the finite plane except for a finite number of singular points interior to a positively oriented simple closed contour C, then

$$\int_C f(z)\,dz = 2\pi i \operatorname*{Res}_{z=0}\left[\frac{1}{z^2} f\left(\frac{1}{z}\right)\right].$$

Thanks